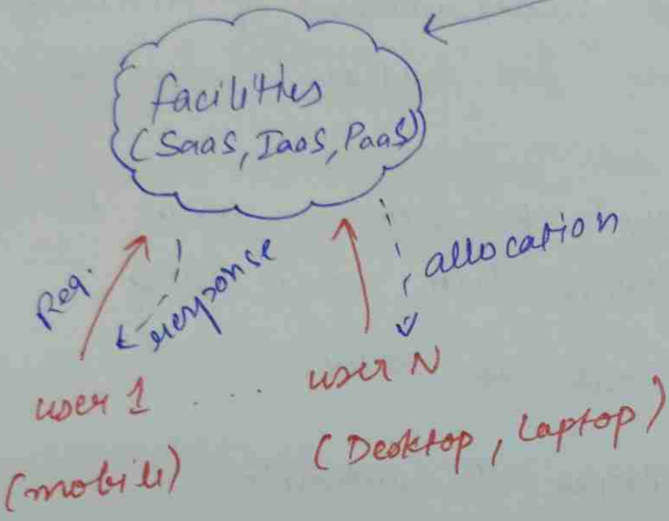
Cloud Computing & virtualization

Cloud : It is something which is present in remote location.

cloud provider offer facilities



- → off-premise computing (Data can be accessed from anywhere)
- → Data Center (Storage H/w)
- → Services

Cloud Computing : It provides us the means by which we can access the applications as utilies over the Internet. {on demand access, shared pool of configurable resources} → as per requirement ~~configure~~

It also refers to manipulating, configuring & accessing the application online.

→ It has the Ability to deliver Computing Services, such as servers, storage, database, networking & Intelligence.

Examples:

① Cloud-Based virtual Desktops:
these help users access their systems & applications by using any device from anywhere.
Eg. VM ware, Horizon Cloud, Virtual windows of Microsoft, virtual desktop providers are Amazon workspace.

② Cloud disaster recovery: this services users to have a backup of their Data when any disaster recovery needs to occur.

Eg. Mozy, Amazon Glacier & Carbonite.

③ Iaas (Infrastructure as a service): It helps businesses to scale their Computer resources up or down whenever needed without any requirement for Capital Expenditure on physical Infrastructure.
Eg. AWS, Google Cloud & Microsoft Azure.

④ Software as a Service (Saas): user can able to access applications hosted in the Cloud, rather than installing & running them on local Devices.

Eg. Salesforce, Dropbox & Microsoft office 365

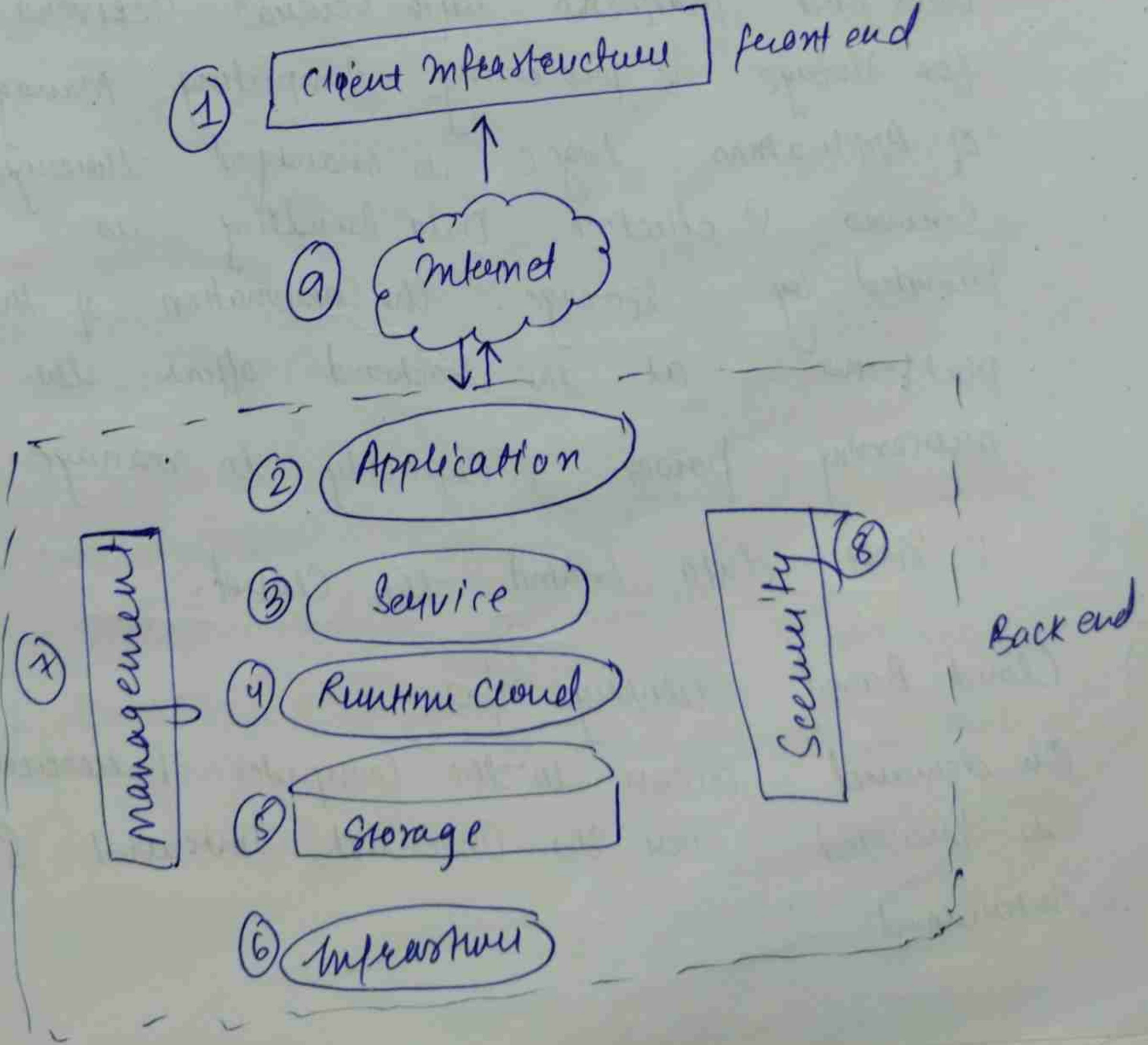
(5) Paas (Platform as a service): this helps organisations with a cloud-Based platform to build, deploy & manage Application.

Eg. Google App engine, Microsoft Azure.

Cloud Computing Architecture:

Consist of Components & sub components

- front end (fat client, thin client)
- Back end (servers, storage)
- Cloud based delivery & n/w (Internet, Intranet, Intercloud)



1. Front End (User Interaction Enhancement)

→ the UI of Cloud Computing consists of 2 sections of clients.

thin

[web Browser facilitating portable & lightweight accessibilities]

Fat

[use many functionalities for offering a strong user Experience]

2. Back end Platform (Cloud Computing Engine)

→ the Core of Cloud computing is made at back end platforms with Several Servers for Storage & processing Computing. Management of Applications logic is managed through Servers & effective Data handling is provided by Storage. the Combination of these platforms at the Backend offers the processing power & Capacity to manage & Store data behind the Cloud.

3. Cloud Based delivery n/w

On demand access to the Computer & resources is provided over the Internet, Intranet & Intercloud.

Internet comes with Global accessibility,
Intranet helps in internal communication of the services within the organization
Intercloud enables interoperability across various cloud services.

Why cloud computing

→ Scalability: Cloud computing services enable organizations to effectively scale up or down their computer capacity to suit changing demands.

→ Accessibility

→ Security

→ Cost-Effectiveness

What is Cloud Computing in AWS?

→ It offers online on-demand access to a large number of computer resources

AWS offers:

- Virtual servers (EC2):
- Storage (S3): store & retrieve
- Database (RDS): set up, run, scale
  : MySQL, postgreSQL & Oracle.
- N/w - virtual n/w
- Security

Characteristics of Cloud Computing:

(1) On demand Self Services: user don't have to communicate with the IT staff.
the self service aspect makes it simple & quick to scale resources as needed.

(2) Broad n/w Access

(3) Resource pooling: to service several client, cloud companies combine their physical & virtual resources.

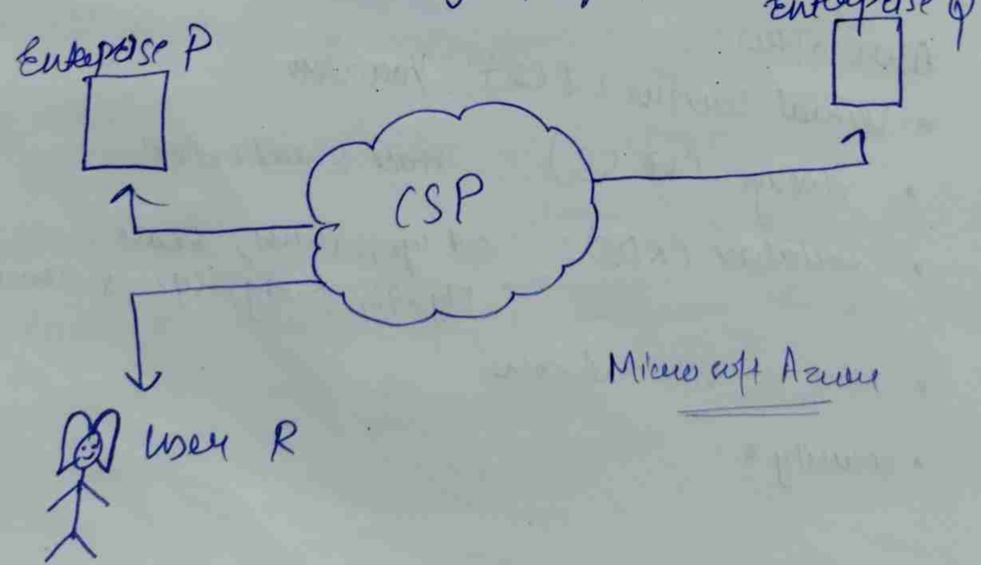
(4) Rapid Elasticity: Scaled up & Down Resources

(5) Measured services:

Types of Cloud Computing: [Cloud Deployment Models]

[1] Public Cloud Computing: It can be access by everyone. It is less secure.

→ the public cloud is one in which cloud infrastructure services are provided over the internet to the General people or major industry Groups.



Microsoft Azure

Eg: Google App Engine. →

Advantages

> Minimal Investment: b/c it is pay per use service, there is no upfront fee.

> No setup cost:

> No maintenance

> Infrastructure management isn't Required

> Dynamic Scalability

Disadvantages:

> less Secure

> less customization

[2] Private Cloud: It is one-on one environment for [Internal Cloud] a single user. there is no need to share your hardware with Anyone else.

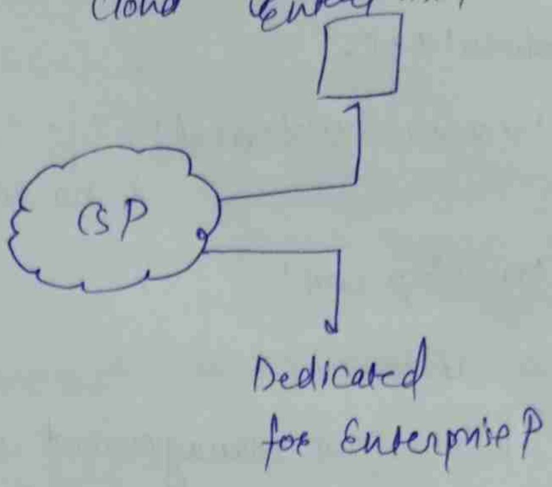
- It has the ability to access systems & services within a Given Border of organization.

- the cloud platform is implemented in a cloud Based Secure environment that is protected by powerful firewalls & under the supervision of an Organizations IT Dept.

VMware, Amazon web ser., Azure, Google cloud, Oracle cloud, Open stack, IBM cloud

- On premise private Cloud

- Externally Hosted Private Cloud



Adv.

- Better Control : Sole owner – IT operate, polices
- Data Security & privacy
- Supports Legacy System
- Customization

dis.adv.

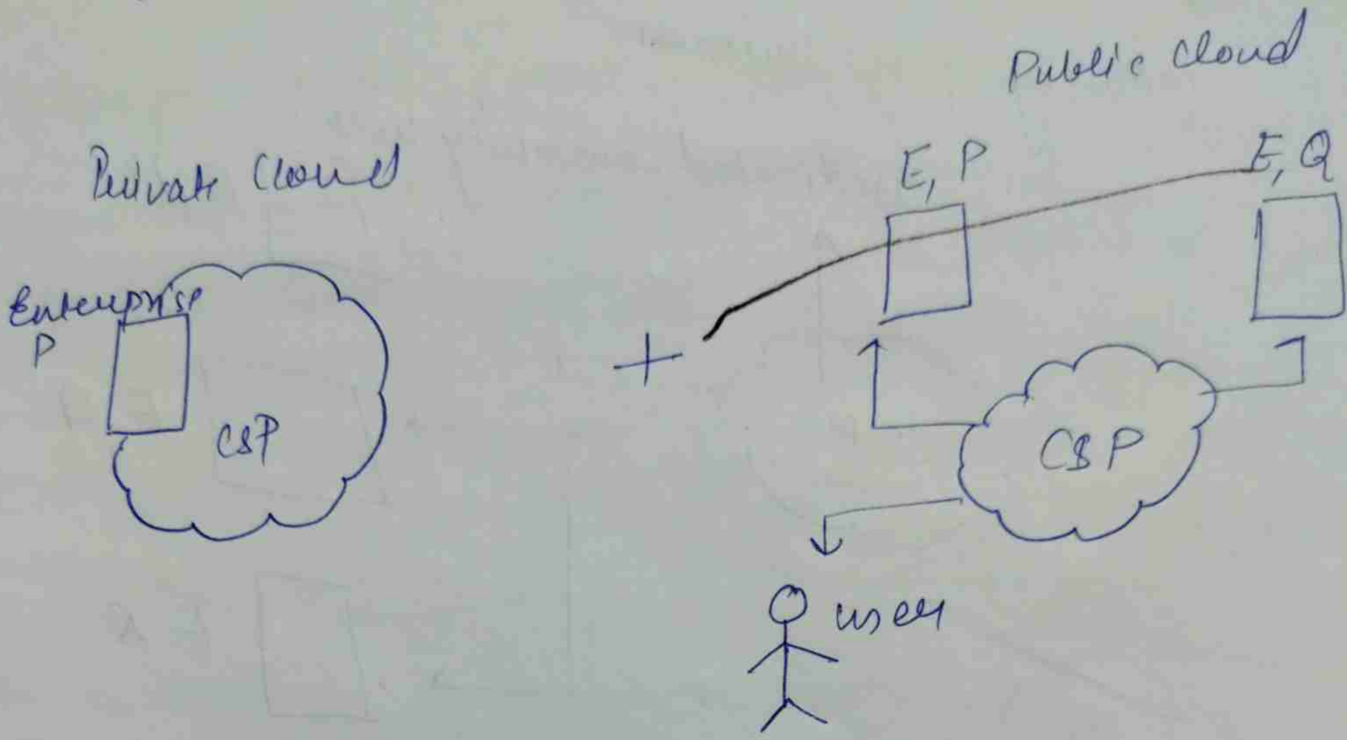
- less Scalable
- Costly

(3) Hybrid Cloud: By bridging the public & private worlds with a layer of proprietary S/w, hybrid Cloud Computing Gives the best of both worlds.

With hybrid Soln, you may host the app in a safe environment while taking Adv. of the public Cloud's Cost savings.

AWS outposts,
Azure stack,
Azure Arc,
Azure VMware Solution,
Nutanix Cloud

Organization can move data & Application b/w different Clouds using Combination of two or more cloud deployment methods, depending on their need.



Adv.

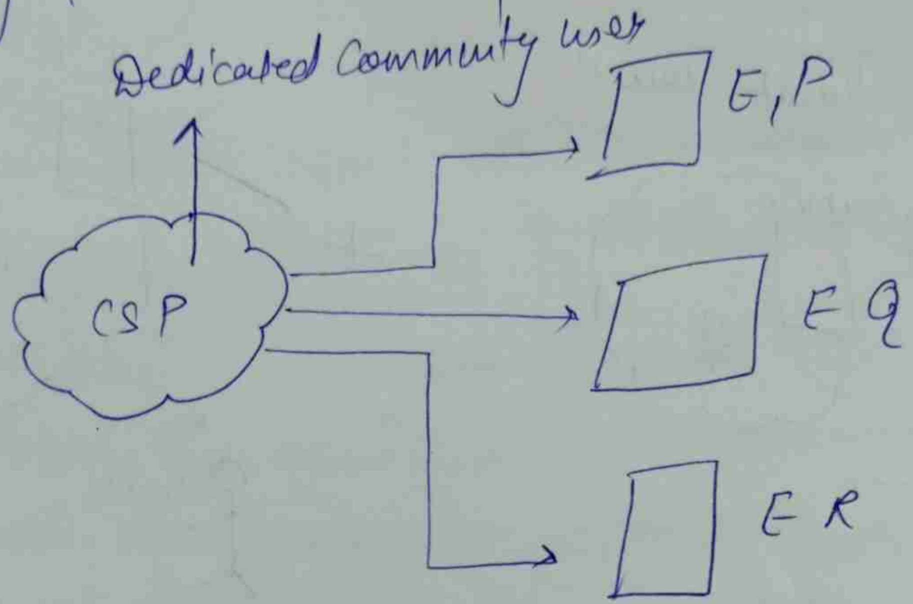
- Flexibility & Control
- Cost
- Security

Disadv.

- Difficulty to manage
- Slow data transmission

[4] Community Cloud: It allows systems & Services to be accessible by a Group of Organizations.

It is Distributed System that is created by integrating the Services of different clouds to address the specific needs of a Community industry, or business.



Adv.

Cost Effective

Security

Shared resources

Collaboration & Data Sharing

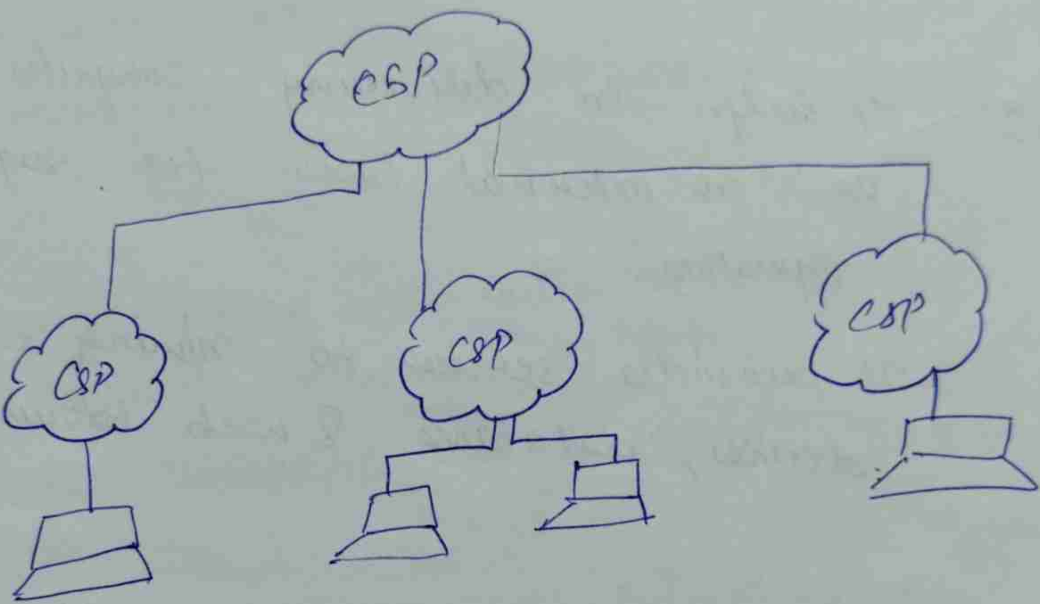
dis. adv.

limited Scalability

Rigid in Customization

[5] Multi-Cloud

It is similar to hybrid cloud, but multi cloud uses many public cloud. Although public cloud providers provide numerous tools to improve the reliability of their services, mishaps still occur.



Adv.

Reduced latency: To reduce latency & improve user experience, you can choose cloud regions & zones that are close to your client

High availability of service:

disadv.

Complex, Security Issues

Models of Cloud Computing

IaaS

PaaS

SaaS

(a) IaaS: It helps in delivering computer infrastructure on an external basis for supporting operation.

It provides services to n/w-ing equipment, devices, databases & web browser.

Adv.

Cost effective, Security, remotely accessible.

Disadv.

- users have to secure their own Data & application
- not accessible in some regions of the world.

(b) Paas: It helps Developers to build applications & services over the Internet by providing them with a platform.

- It helps in maintaining control over their business Applications.

Adv.

- convenient & simple
- manageable

dis. adv.

- limited control over infrastructure as they have less control over the env. & not customizable
- high Dependence on the provider.

(c) Saas: It is the work of Delivering Services & applications over the Internet.

Saas application are called web-based s/w & hosted s/w.

Adv. - can be accessible app data from anywhere on the Internet
- Easy accessible to feature & services

disadv.

- limited Customization
- little Control over Data
- It require stable Internet

Q How to Secure data while transferring?

II Classic Data Center (CDC)

It is a facility containing physical IT resources including Computer, n/w & storage.

A data Center facility Enables an Organization to assemble its resources & Infrastructure for data processing storage & communication

Including!

- systems for storing, sharing, accessing & processing data across the Organization.
- physical Infrastructure to support data processing & data Communication
- Utilities such as Cooling, electricity, n/w access, UPS (uninterruptible power supplies)

Main Components of Data Center :-

① Calculation

② Enterprise data storage

③ N/w

A modern data center concentrates an organization's data systems in a well-protected physical infrastructure which includes :-

- Server (program which provide services)
- storage subsystems
- n/w switches, routers & firewall
- cabling
- physical racks for organizing.

Key Characteristics of CDC

- Physical Infrastructure : the organization owns & maintains the physical H/w such as server, storage systems & n/w devices.

Scalability: involves purchasing & installing additional h/w which is time consuming.

Management: regular updates, monitoring

Control & Cost structure.

Eg: Email, ERP (Enterprise Resource planing)
Data warehouse (DW)

* Compute: A resource that runs applications with the help of underlying computing components.

Physical (h/w) + logical (s/w & protocol)

CPU Memory I/O

Memory is used to store Data either persistently or temporarily.

RAM (Random access memory)

ROM (Read only memory)

RAM is used to store data that the Computer needs to boot & operate.
RAM is faster than ROM as the data stored in it can be accessed & modified in any order, while data stored in ROM can only read.

1° memory (RAM & ROM)
2° memory (Hard drive, CD)

(volatile) At data lost
RAM
- SRAM (Static) [Computing devices & microprocessor]
- DRAM (Dynamic) [capacitors, has small life span]

(non-volatile)
ROM
- PROM (programmable)
- EPROM (Erasable program)
- EEPROM (Electrically)

Example of Computer system
- laptops/Desktops
- Blade Server
- Complex cluster of servers
- mainframe.

Blade Servers: A more compact form of Servers that can be housed in a single chassis. They offer high-density computing & are often used in Environments where space is at a premium.

[Read: Tower Server, Rack Server]

→ ~~The~~ Blade Servers were developed in response to the growing need of a computing power that Doesn't consume a large floor space. They bring down the Cost & Complexity in the Datacenter.

These servers consolidate power- & system-level functions into single, integrated Chassis & enable the addition of server modules as hot-pluggable components

Blade Server technology greatly ~~ses~~ server density, lowers power & cooling costs, eases server Expansion & simplifies ~~...~~ datacenter management

Server Clustering:

Multiple servers (nodes) are brought together in a cluster to improve availability & performance.

→ When failure occur on one node in a cluster, resources & workload are redirected to another node.

→ Exchange heartbeat is a checkup mechanism b/w two nodes.
- To see wheather a node is up & running
- A failover is initiated, if heartbeat fail.

Storage: It is resource that stores Data persistently for subsequent use,

Eg: magnetic, optical, solid state media,
- magnetic → Disk Drive
- optical → CD
- solid state → Flash Drive

Tape Drive ~~optical~~

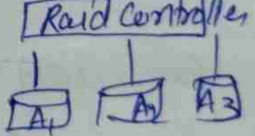
RAID (Redundant Array of Independent Disk)

A technology which utilizes multiple disk drives as a set to provide protection, capacity &/or performance benefits.

- Overcomes limitations of Disk Drive
- Improves storage system performance
- Technique → striping
  → mirroring
  → parity

Striping: It is a technique of spreading data across multiple drives in order to use the drives in parallel.
All read-write work simultaneously this allows more data to be processed in a shorter time.

Performance ↑



Mirroring: It is a technique where data is stored on two different disk drives, yielding two copies of data. In the event of one drive failure the data is intact on the surviving drive & the controller continues to service the

ABGL H

Comput systems data req. from the surviving disk of the mirrored pair.

- improves read performance b/c read req. are serviced by both Disk.
- Expensive, b/c it involves duplication of Data
- Nested Raid

Parity: It is a method of protecting striped data from Disk failure without Cost of mirroring.

An additional Disk Drive is added in the strip set to hold parity, a mathematical construct that allows re-creation of the missing Data.

* Intelligent storage system:

> It is a raid array highly optimized for I/O processing
> It refers to a data storage Solution that utilizes advanced technologies like AI, ML & automation to optimize the storage, management & retrieval Data.
> large amount of cache For improving I/O performance

Example - NetApp ONTAP, Dell EMC PowerStore, IBM Spectrum Storage.

Key features & Components of an IDS.

(1) Automation - SSD, HDD

(2) AI & Machin Learning Integration
- Predictive Analytics & Anomaly Detection

Components of IDS

I Storage H/W

(a) SSD (Solid State Drives) - high-speed storage Devices that offer faster Data access times compared to traditional HDD.
SSD are often used for storing frequently accessed Data.

(b) Hard disk Drive (HDD) - Cost effective storage device with larger capacities, typically used for less frequently accessed Data.

(c) NAS (n/w attached storage).

(d) Storage area n/w (SAN)

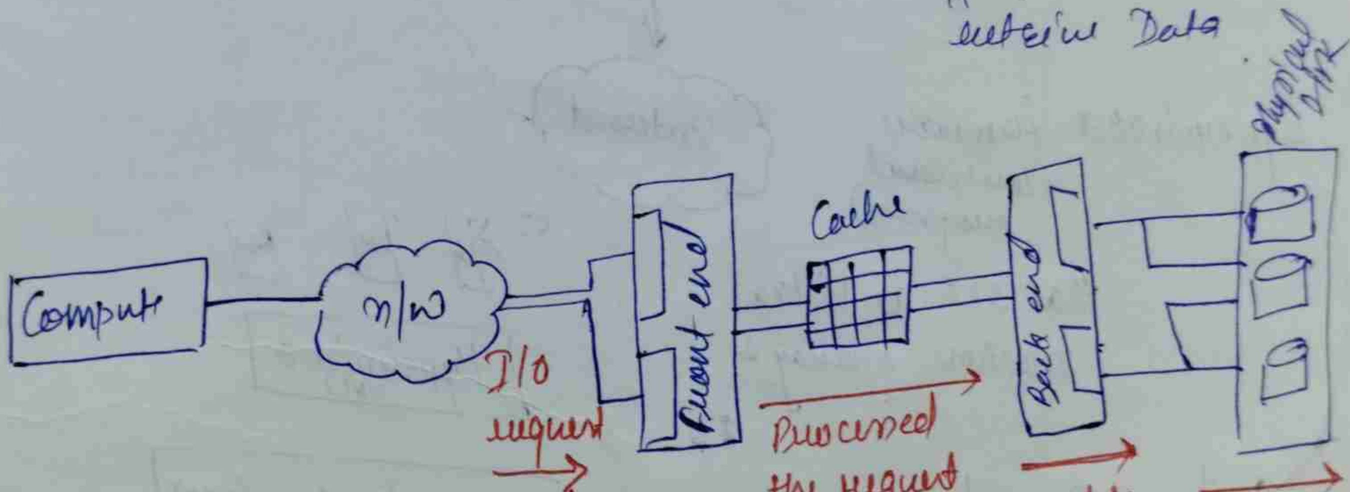
Component
- Front end
- Cache
- Back-end
- Physical Disk

* Cache: It is a high speed data storage layer that stores a subset of Data, typically transient in nature so that future req. for that Data can be served faster than if the Data had to be fetched from the IO storage location.

Caches are used to reduce the time it takes to access data & to reduce the workload on the underlying data storage or processing system.

Adv. - Performance Enhancement, Reduced latency, load Balancing

↳ By storing Data closer to the point of use, caches reduce the time it takes to retrieve Data



Processed the request through cache

enable to store the Data in Disk

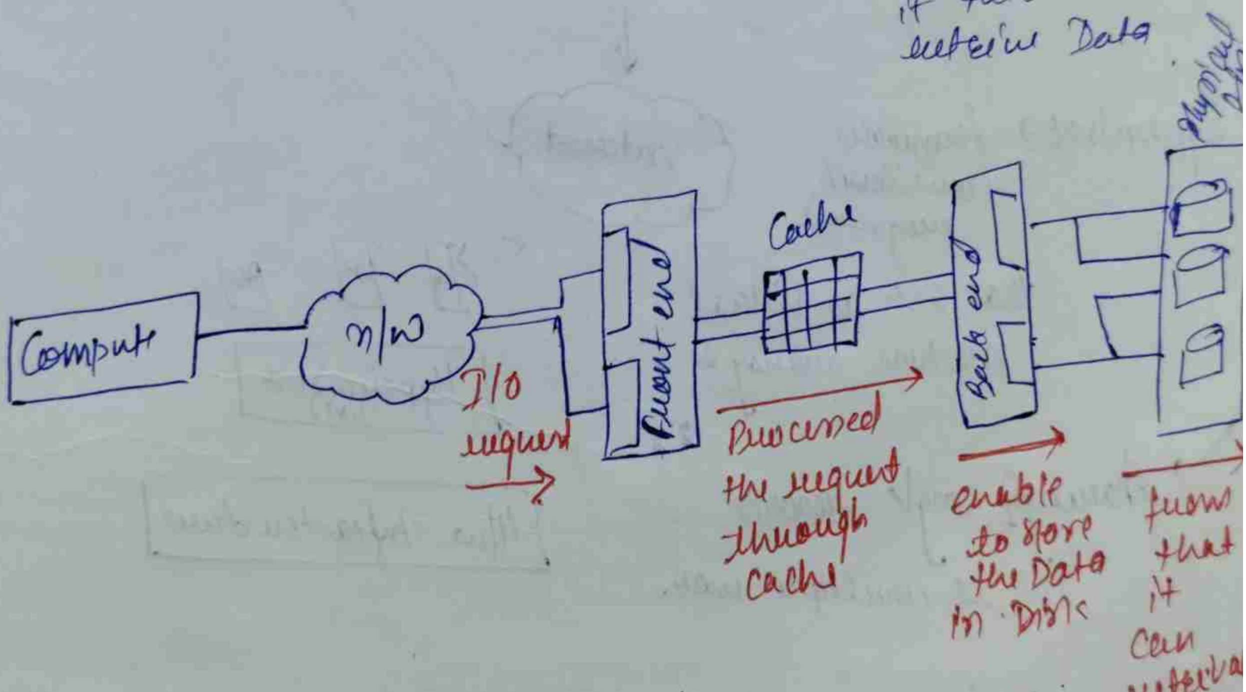
from that it can retrieval of Data from PD.

* A read req. can be serviced directly from cache if the requested Data is found in Cache.

* Cache: It is a high speed data storage layer that stores a subset of Data, typically transient in nature so that future req. for that Data can be served faster than if the Data had to be fetched from the 1° storage location.

Caches are used to reduce the time it takes to access data & to reduce the workload on the underlying data storage or processing system.

Adv. - Performance Enhancement, Reduced latency, load Balancing
↳ By storing Data closer to the point of use, caches reduce the time it takes to retrieve Data.

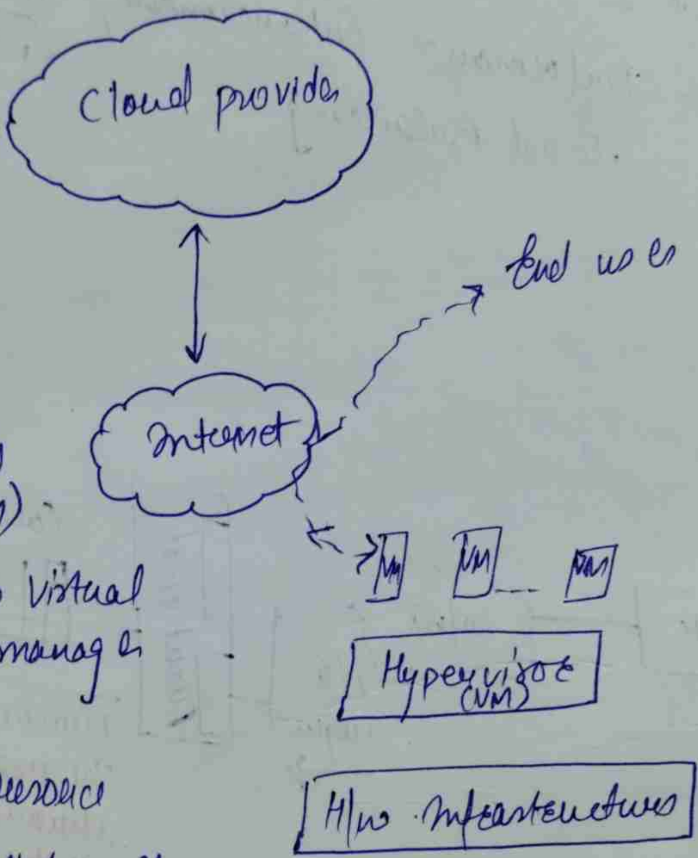


* A read req. can be serviced directly from cache if the requested Data is found in Cache.

Cloud Computing Infrastructure

↳ Storage devices
↳ Servers
↳ S/w } • cloud mgmt
       } • for users

↳ VM



Hypervisor → firmware (low level program) that act as virtual machine manager

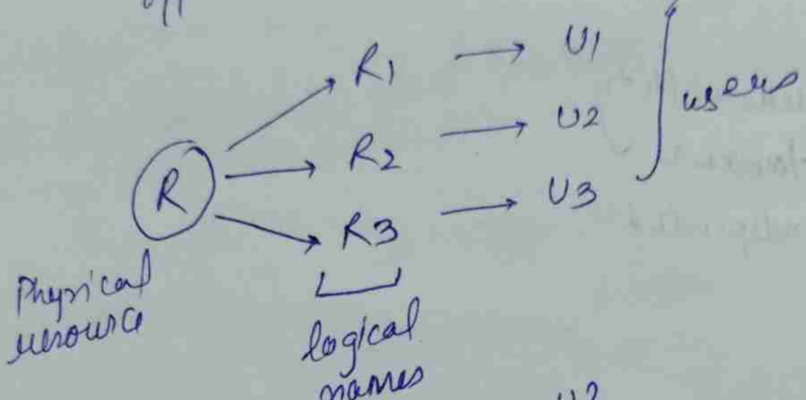
→ sharing of single resource to multiple user.

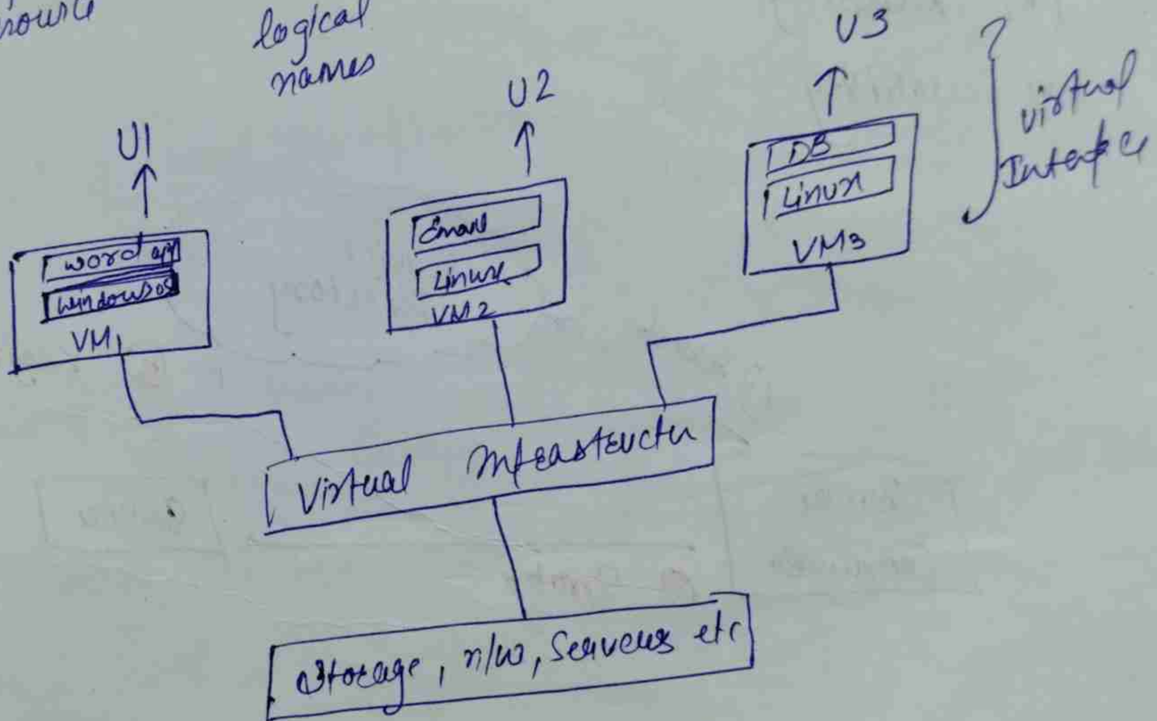
Constraints/ Tasks. ① Scalability
② Transparency
③ Monitoring
④ Security

Cloud Computing technologies

→ Virtualization
→ Service oriented architecture
→ Grid Computing
→ utility computing.

(i) Virtualization: Allows to share single physical instance of an application/resource among multiple users.



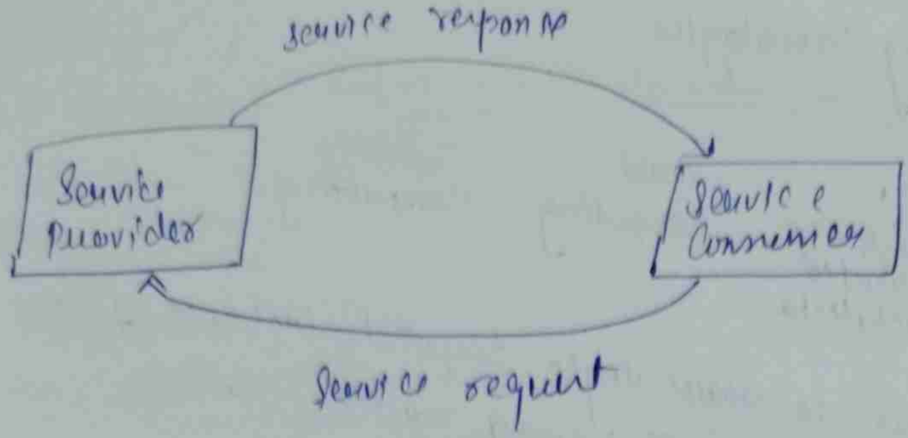


(ii) SOA (Service Oriented Architecture):

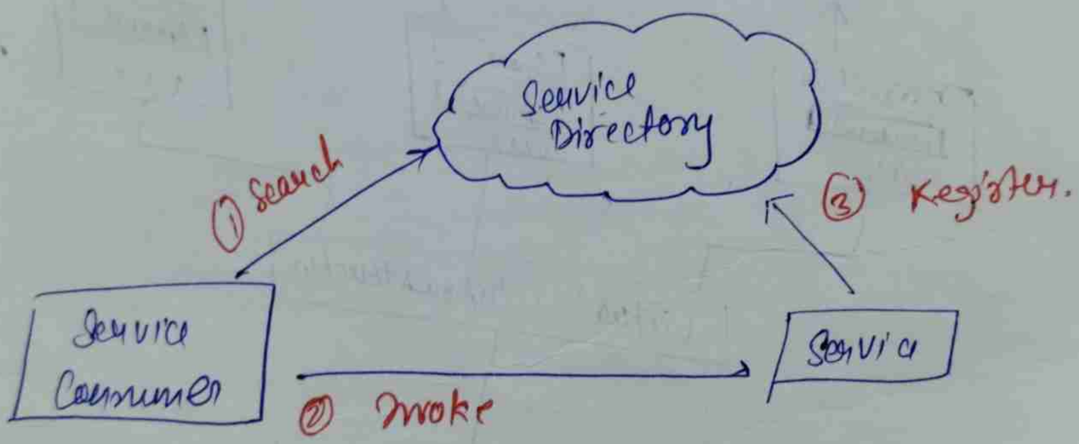
Services are provided to form application through internet

→ allows to use Application as a service for other Application

→ allows to exchange of Data b/w various Application



Adv.
(i) Service reusability
(ii) Easy maintenance
(iii) platform independent
(iv) Reliability
(v) Scalability



(iii) Grid computing
↳ Distributed Computing in which a Group of Computers from multiple locations are connected with Each other.

→ Computer resources can be heterogenous & can be located Anywhere.

Utility Computing: It is a model of computing where computing resources, such as processing power, storage, & applications, are provided to users as a metered Service, similar to how utilities like electricity or water are supplied.

Ex - Scientific Research, Business Application

Cloud Management in Cloud Computing

It is maintaining & controlling the Cloud Services & resources be it public, private or hybrid.

Aspects of cloud management → load balancing, performance, storage, backups, capacity, deployment etc.

Need of cloud Management → Cloud is nowadays preferred by huge organizations as their 1° data storage. A small downtime or an error can cause a Great deal of loss & inconvenience for the Organization. So as to design, handle & maintain a Cloud Computing Service specific members are responsible who make sure things work out as supposed & all arising issues are Addressed.

Tasks of Cloud management:

- **Auditing System Backup** ← (audit the backups time to time to ensure the restoration of randomly selected files of different users)
- **Flow of data in the system** → (the managers are responsible for designing a DFD that shows how the data is supposed to flow throughout the organization).
- **Vendor lock in** → (the managers should know how to move their data from server to another in case the organization decides to switch the providers)
- **Knowing providers security procedures** → (Encryption policy & Employee screening)
- **Monitoring the capacity scaling abilities** → (Scaling & planning)
- **Monitoring audit log** → (In order to identify errors in the system, logs are audited by the managers on a regular basis.)
- **Solution testing & validation** → (It is necessary to test the cloud services & verify the results & for ~~free~~ error free s/m)

Cloud Storage

It is a virtual locker where we can remotely store any Data. When we upload a file to a cloud based server like Google Drive, OneDrive, or iCloud that file gets copied over the Internet into a Data server that is cloud based actual physical space where companies store files on multiple hard Drives.

features of Cloud Storage System

→ It has a Greater Availability of Resources
→ Easy maintenance as
→ It has a large n/w Access.
→ It has an automatic System.
→ security

there are 3 types of storage systems

> Block - Based storage system
> file - Based Storage System
> Object Based storage system

1. Block - Based Storage System :

hard Drives are block-based storage system. Your OS like windows or linux actually sees a hard ~~drive~~. disk drive.

2. File-Based Storage System

In this, you are actually connecting through a n/w interface Card (NIC). You are going over a n/w & then you can access the n/w - attached storage server (NAS).

NAS Devices are file based storage system.

3. Object Based Storage System -

In this, a user uploads object using a web browser & uploads an object to a Container i.e., Object Storage Container. This uses the HTTP protocols with the rest of the API (for example, Get, Put, Post, Select, Delete.)

Cloud Storage Architecture :-

- The Cloud Storage Architecture Consists of several Distributed resources, but still functions as one, either in a Cloud Architecture of federated or cooperative storage.
- Durable through the manufacture of copies of versions
- ultimately, it is usually Compatible with Data replication advantages.
- Companies just need to pay for the storage they actually use, normally an average of a month's consumption this doesn't indicated that cloud storage is less costly, but rather that operating costs are incurred rather than Capital Expenses.

Advantages

Scalability, flexibility, Simple Data Migration (removement & old Data when required & eliminated disruptive data migrations), Recovery.

Disadv

→ Data Centers require electricity & proper internet facility to operate their work, failing which system will not work properly.

→ Support for Cloud storage isn't the best, especially if you are using a free version of a Cloud provider.

→ When you use a Cloud provider, your Data is no longer on your physical storage.

Q How Cloud storage Benefits Business.
Q How cloud storage actually works. (GFG)
Q Read & Real world Applications of Cloud Computing.

Storage networking Technologies

Virtualization

It is used to create a virtual version of an underlying service with the help of virtualization, multiple OS & Application can run on same machine & the same hardware at the same time, increasing the utilization & flexibility of H/w.

It was initially Developed during the mainframe era. large scale Computation & data processing tasks.

→ It delivers IAAS solution for CC.

Host machine

the machine on which the VM is Going to B uilt

Guest machine.

the VM is is referred as a Guest machine.

Benefits of virtualization

→ More flexible & efficient allocation of resources
→ Enhance Development productivity
→ It lowers the Cost of IT infrastructure.
→ Remote access & rapid Scalability.
→ High availability & Disaster recovery.

Drawback of virtualization

→ High Initial investment: Cloud have a very high initial investment, but it is also true that it will help in reducing the Cost of Companies.

→ Learning new infrastructure :- As the Companies shifted from servers to Cloud, it requires highly skilled Staff who have skills to work with Cloud easily, & for this, you have to hire new staff or provide training to current staff.

→ Risk of Data : hosting Data on third party resources can lead to putting the Data at risk, it has the chance of Getting attacked by any hacker or cracker very easily.

Characteristics of virtualization

(1) Increased security:

(2) Managed Execution - In particular, sharing, aggregation, emulation & isolation are the most relevant features

(3) Sharing: virtualization allows the creation of a separate computing environment with the same host.

(4) Aggregation - Share physical resources among several Guests

Types of virtualization

- Application
- n/w
- Desktop
- Storage
- Server
- Data

4/9/24

Deadline: 06-9-2024

Assignment - I

Q. Describe the key elements of a CDC. What are the core components that make up a CDC.

Q. What are the key requirements of Data Center? Explain the significance of each Requirement.

Q. List & Explain the different types of Applications commonly deployed in a CDC. What are the key I/O characteristics of these application.

Q. What is a DBMS? How does a DBMS optimize data storage & retrieval?

Q. What are the physical & logical components of a compute system? Provide example.

Q. Explain the concept of ~~Zoning~~ Server Clustering? How does it enhance the Availability & performance of a system?

Q. What is RAID technology? Discuss the different RAID techniques & their Benefits.

Q. What is an ~~ISS~~ Intelligent Storage system? How does it optimize I/O processing.

Q. Explain the Concept of Zoning in Fibre Channel SANs. What are the different types of zoning & how do they function?

(1) Application virtualization: It helps a user to have remote access to an Application from a server. The server stores all personal information & other characteristics of the Application but can still run on a local workstation through the Internet.

Ex- hosted Application & packaged application.

(2) N/w virtualization: The ability to run multiple virtual networks with each having a separate Control & Data plan. It co-exists together on top of one physical n/w. It can be managed by individual parties that are potentially confidential to each other.

Network virtualization provides a facility to create & provision virtual n/w, logical switches, router, firewalls.

(3) Desktop virtualization: It allow the users OS to be remotely stored on a server in the Data Center. It allow the user to Access their desktop virtually, from any location by a different machine.

The main benefits of Desktop virtualization are user mobility, portability, & easy management of s/w installation, updates & patches.

(4) Storage virtualization: It is an array of servers that are managed by a virtual storage system.

the servers aren't aware of ~~servers~~ exactly where their data is stored & instead function more like worker bees in a hive.

Server virtualization: this is a kind of virtualization in which the masking of server resources takes place. Here the central server (physical server) is Divided into multiple Different virtual servers by changing the identify no. & processor.

Assignment - 2

Q. How does fibre channel over Ethernet (FCoE) consolidate n/w traffic? what are the key benefits of FCoE in a CDC.

Q. Describe the key component of an FCoE n/w. what roles do the converged n/w Adapter (CNA) & FCoE switch play?

Q. How does IP-SAN differ from traditional FC-SAN? highlight the adv. of using IP as a storage transport?

Q. Explain the diff. b/w iSCSI & FCIP in IP-SAN;

Q. what are the two topologies used in iSCSI implementation?

Q. what is the role of an iSCSI gateway in a bridged iSCSI topology?

Q Describe the key benefits using FCoE over traditional
2 storage communication technologies.

Zoning in Fibre Channel SANs :-

It is a method used in Fibre Channel SANs to control which devices can communicate with each other.

There are three types of zoning!

① WWN zoning: Based on world wide name (unique identifiers for devices).

② Port zoning: Based on the physical port on the switch to which devices are connected.

③ Mixed zoning: Combines both

DL → Tensorflow, keras, Pytorch

ML → numpy, Pandas, Scikit, matplotlib

NLP → NLTK

Compute Virtualization

- It is a technique of masking or abstracting the physical H/w from the OS & enabling multiple OS to run concurrently on a single or clustered physical machine.
- this technique encapsulates an OS & an Application into portable VM.
- A VM is a logical entity that looks & behaves like a physical machine. Each OS runs on its own VM.

In compute virtualization, A virtualization layer resides b/w the hardware & VM. the virtualization layer is also k/a hypervisor.

the hypervisor provides a standarized H/w resource (example: CPU, memory, n/w)

Hypervisor:

It interacts Directly with the physical resources of the x86 based compute system.

It has two components:

1. kernel
2. virtual machine monitor (VMM)

Kernel → It provides the functionality like process creation, file system management & process scheduling.

→ It is designed to specifically support multiple VM & to provide core functionalities, such as resource scheduling, I/O stacks.

VMM → It is responsible for actually executing commands on the CPUs & performing Binary Translation (BT).

Hypervisor types

Type 1 Bare Metal Hypervisor
→ It is an OS, installs & run on x86 bare metal h/w. It requires certified H/w. It is more efficient than a hosted hypervisor.

Type 2 Hosted Hypervisor
→ It install & run as an Application. It relies on OS running on physical machine for device support & physical resource management.

Benefit of Compute virtualization

- Server Consolidation – Compute virtualization enables running multiple VM on a physical server. this reduces the requirement for physical server.
- Isolation: while VM can share the physical resources of a physical machine, they remain Completely isolated from each other as if they were separate physical machines.
  Example: there are four virtual machines on a single physical machine & one of the virtual machines crashes, the other three VM remain unaffected.
- Encapsulation: A VM is a package that Contains a Complete set of virtual H/w resources, an OS & Applications. Encapsulation makes VM portable & easy to manage. Ex. a VM can be moved & Copied from one location to another just like file.

- H/w Independence: A VM can be configured with virtual component such as CPU, memory, n/w card & SCSI controller that are completely independent of the underlying physical H/w.
- Reduced Cost: Compute virtualization reduces the following direct costs: space, power & cooling, H/w & annual maintenance.

Requirements: x86 H/w Virtualization

An OS is designed to run on a bare metal h/w & fully own the H/w.

H/w virtualization allows multiple OS to run concurrently on a host computer. It utilizes special CPU instructions to improve performance & isolation of VM.

Key technologies: Intel VT-x (Intel virtualization technology)
↳ Enables the creation of virtual environments on Intel processor.

! AMD-V (AMD virtualization): similar function.

Bios utility
↳ named as SVM mode (secure VM)

Windows: CMD → run systeminfo
→ virtualization Enabled in firmware
→ ensure "Yes".

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Linux - egrep -c '(vmx|svm)' /proc/cpuinfo
(to check if virtualization extension are present)

(3). Bios/UEFI setting:

F2, F10, DEL (system boot)

Intel VT-x }
AMD-V } enable.

Full virtualization

It is a type of virtualization where VM is completely isolated from the host OS & other VMs.
It provides a way to run multiple OS on a single physical machine without requiring modification to the Guest OS.
the hypervisor (VMM) provides an Abstraction layer between the Guest OS & physical H/w.

Key features
① Complete Isolation: Each VM runs independently of other & the host OS, providing strong isolation b/w different virtual Environment.

② No modification required

③ Hardware Abstraction: the H/w provides virtualized H/w (CPU, memory, storage) to the Guest OS.

the Guest OS is unaware that it is running in a virtualized Environment.

Disadv - Performance Overhead
- Complexity - Managing & Configuring multiple VMs can be Complex, especially in large Scale Env.

Paravirtualization

It is a type of virtualization where the Guest OS are aware that they are running in a virtualized Environment & are modified to work with the hypervisor.
- this approach can offer performance benefits over full virtualization by reducing the overhead associated c̄ emulating H/w.

Key features:

① Guest OS Modification
It requires changes to the Guest OS, which may not be feasible for all OS types or Application. Not all OS support paravirtualization.

② Efficient Resource management
P.V can reduce the Overhead typically associated with virtualization by streamlining communication b/w the Guest OS & the hypervisor.

this can lead to better performance for certain types of operations, as the hypervisor can optimize resource allocation more effectively.

Reduced Emulation Overhead: Since the Guest OS is modified to work directly with the hypervisor, there is less need for the hypervisor to emulate hardware. this can lead to increased efficiency & performance improvements, especially in high throughput scenarios.

Adv. Improved performance
Lower Overhead
Better Scalability

Disad. Compatibility Issue

Example: Technologies like XEN (an open source hypervisor), VM ware, Hyper-V

~~Virtual Machine Files~~

~~virtual BIOS file: It stores the state of VM BIOS~~
~~virtual swap file:~~